REVIEW · NUTRITION SCIENCE

Are seed oils really bad for you?

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ABSTRACT

Ordinary use of seed oils is best understood as a source of unsaturated fat, not as a general toxic exposure; when these oils replace fats richer in saturated or trans fatty acids, the clearest effect is an improvement in atherogenic lipoproteins. Randomized evidence for clinical events is older and less decisive, although one synthesis of 13,614 participants estimated a coronary-event risk ratio of 0.81, while prospective intake and biomarker studies are predominantly neutral or favorable. The strongest contrary evidence comes from recovered data in two mid-century trials: cholesterol lowering did not reduce mortality in one, and mortality increased in a small secondary-prevention trial, but incomplete records, unusual interventions, and limited generalizability prevent those results from defining the whole category. The conclusion does not extend to repeatedly overheated oil or make foods healthy merely because they contain seed oil. A practical evidence-based position is to prefer mostly unsaturated oils, avoid repeated frying and smoking, and evaluate the complete food and its replacement.

01 Introduction

“Seed oils” commonly refers to oils pressed or extracted from soybean, corn, rapeseed or canola, sunflower, safflower, cottonseed, grapeseed, rice bran, and related seeds. The category is prominent in public debate because many of these oils supply linoleic acid, the principal dietary omega-6 polyunsaturated fatty acid. Claims of harm usually combine several propositions: linoleic acid is said to amplify inflammation, its double bonds are said to generate toxic oxidation products, and the historical rise in industrial oils is said to explain chronic disease. Yet the oils differ in fatty-acid composition, appear in very different foods, and can be consumed fresh, baked, sautéed, or repeatedly deep-fried. A category-level verdict is therefore biologically underspecified.[1,2,3]

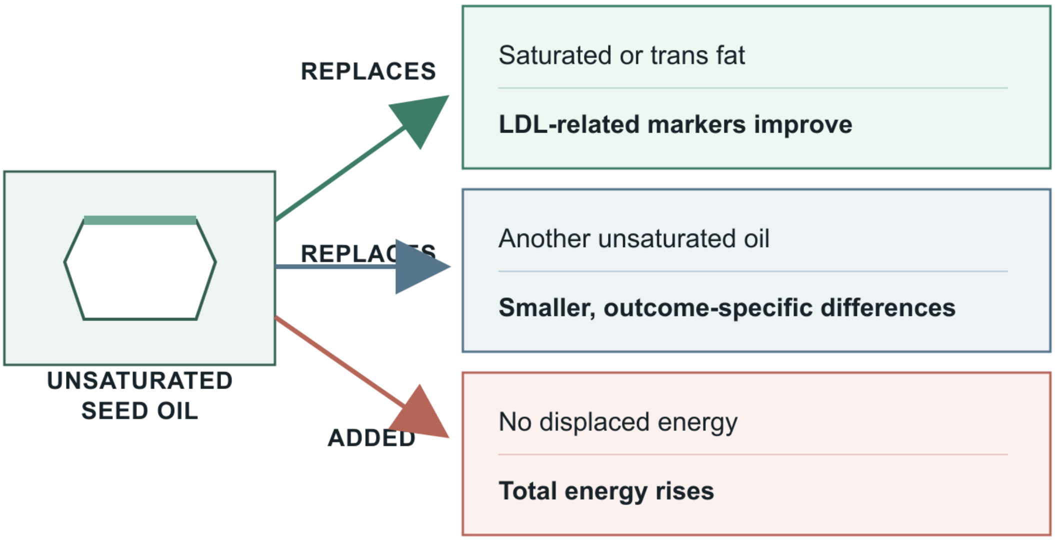
The clinically relevant question is not whether an oil has a biochemical pathway capable of harm. Almost every nutrient does. It is whether ordinary consumption, at realistic doses and preparations, worsens validated human outcomes compared with the food or nutrient it displaces. Controlled feeding can answer short-term causal questions about blood lipids and glucose; long trials can address events but are scarce; prospective cohorts add scale and duration while retaining confounding; and heating studies define a separate exposure boundary. This review therefore asks how oil composition, replacement nutrient, food matrix, and heating history jointly determine the answer.

02 Seed oils are a heterogeneous category, and the replacement defines the contrast

Conventional sunflower, corn, and soybean oils are usually rich in linoleic acid, whereas canola contains more oleic acid and some alpha-linolenic acid; high-oleic sunflower and soybean oils are chemically different from their conventional namesakes. Reviews that distinguish these formulations find that their oxidative stability and lipid effects cannot be inferred from the words “seed oil” alone.[1,4] Even within one species, breeding, refining, antioxidant content, and storage alter the product. The category is useful for grocery shopping, but it is a poor exposure variable for causal inference.

Replacement is the second necessary coordinate. A spoonful of oil added to an unchanged diet adds energy; the same spoonful replacing butter, partially hydrogenated shortening, or refined carbohydrate creates different metabolic contrasts. Prospective substitution models associate replacing animal foods with plant foods with modestly lower cardiometabolic and mortality risk, but the estimates are tied to the displaced food rather than to an intrinsic property of plants.[5] Quantitative models of replacing partially hydrogenated oils likewise predict different coronary effects for saturated, monounsaturated, and polyunsaturated alternatives.[6]

This dependence is visible in both cohorts and trials. Dietary analyses estimate coronary or stroke risk when linoleic acid replaces saturated fat, trans fat, or carbohydrate; they do not estimate the effect of a free-standing molecule.[7,8] Controlled trials make the exchange explicit by holding calories near constant and changing dietary fats.[9,10] As Figure 1 summarizes, the same oil can look favorable, neutral, or calorically adverse under different counterfactuals. The primary question is therefore “seed oil instead of what?”



LDL = low-density lipoprotein

Figure 1. The replacement nutrient determines the causal question. The central balance separates isocaloric substitution from addition: lipid benefit is clearest when unsaturated oil displaces saturated or trans fat, whereas comparisons among unsaturated oils are smaller and adding oil changes energy intake. Arrows summarize directions rather than universal effect sizes.[6,7,9,5]

03 LDL-related lipoproteins provide the clearest randomized signal

When lower-saturated seed oils replace saturated fats, reductions in low-density lipoprotein (LDL) cholesterol and related atherogenic markers are reproducible. Replacing dairy fat with rapeseed oil rapidly improved hyperlipidaemia in a randomized trial.[11] In moderately hypercholesterolaemic adults, plant oils with different fatty-acid profiles changed cardiovascular risk factors in directions predicted by their composition.[9] A controlled feeding trial found a more favorable lipoprotein profile with corn oil than with extra-virgin olive oil, illustrating that an oil can outperform another on LDL without being superior on every health dimension.[12]

The comparator explains much of the apparent conflict. A meta-analysis found that palm oil, which is relatively saturated, raised LDL cholesterol compared with vegetable oils low in saturated fat.[13] By contrast, a synthesis comparing canola and olive oil found relatively small differences across lipid outcomes.[14] Mid-oleic sunflower oil and olive oil both fit a cholesterol-lowering diet, with their differing unsaturated-fat balance shaping secondary lipid measures.[15] These are not contradictions; they are distinct exchanges.

Walnut and vegetable-oil feeding experiments sharpen the point while adding a food-matrix caution. In an exploratory crossover analysis of 34 adults at cardiovascular risk, the walnut diet lowered LDL cholesterol by 6.9 mg/dL, or 6.5%, relative to an oleic-acid comparison, without increasing lipoprotein(a).[16] In the parent trial, all low-saturated-fat diets improved serum lipids, but whole walnuts lowered central diastolic pressure more than the matched oleic-acid replacement; systolic pressure and arterial stiffness were unchanged.[17] The fatty-acid exchange explains part of the response, but fiber, polyphenols, and food structure can add effects.

Not every surrogate responds. In a double-blind crossover substudy of 31 adults with central adiposity, conventional and high-oleic canola diets produced no difference in flow-mediated dilation after six weeks; mean values were 6.32%, 6.96%, and 6.41% across diets, with P=0.81.[18] The larger DIVAS trial similarly found no vascular-function benefit despite favorable lipid biomarkers, E-selectin, and blood pressure when unsaturated fats replaced saturated fats.[19] Thus LDL-related measures are the clearest causal signal, not a guarantee that every intermediate endpoint changes over weeks.

Table 1 places the main controlled contrasts side by side. It also shows why claims about “seed oils” cannot be read from one comparison.

CONTRAST	LDL	BOUNDARY
Rapeseed for dairy fat[11]	Lower	Events untested
Canola variants[18]	Lower	Vascular effect null
Walnuts or oils[17,16]	Lower	Matrix mattered

04 Clinical cardiovascular outcomes are compatible with modest benefit but remain less certain

Long-term randomized evidence is much thinner than the lipid literature. A meta-analysis of eight trials, including 13,614 participants and 1,042 coronary events, estimated a coronary-event risk ratio of 0.81 (95% confidence interval (CI) 0.70–0.95) when polyunsaturated fat replaced saturated fat; each 5% of energy exchanged corresponded to roughly 10% lower risk.[20] That result supports benefit, but the contributing trials were conducted decades ago, some used oils containing both omega-6 and omega-3 fatty acids, and several had design limitations that would not meet modern standards.

Broader systematic reviews are more guarded. A Cochrane synthesis of omega-6 interventions found little evidence for major effects on mortality and uncertainty around cardiovascular events, while a total-PUFA review suggested possible reductions in coronary events with low or moderate certainty.[21,22] A primary-prevention review likewise found insufficient precision for firm outcome claims.[23] A later synthesis combining randomized trials and Mendelian randomization did not identify a large, consistent cardiovascular hazard, but method-dependent estimates remained heterogeneous.[24] The defensible inference is possible modest benefit, not certainty.

Two recovered-data analyses drive much of the continuing controversy. In the Minnesota Coronary Experiment, 9,423 institutionalized participants were randomized, but only 2,355 remained on the assigned diet for at least one year. Serum cholesterol fell 13.8% in the vegetable-oil group versus 1.0% in controls, yet mortality did not improve; the associated updated coronary-mortality meta-analysis gave a hazard ratio of 1.13 (95% CI 0.83–1.54).[25] This is evidence that lipid improvement within one compromised historical trial did not translate into observed survival benefit. It is not evidence that LDL has no causal relevance or that modern ordinary exposure is harmful.

The Sydney Diet Heart Study is the sharper adverse result. Among 458 men with a recent coronary event, a safflower-oil intervention was associated with all-cause mortality of 17.6% versus 11.8% and a hazard ratio of 1.62 (95% CI 1.00–2.64); cardiovascular and coronary mortality were also higher.[26] However, records were incomplete, the original protocol was unavailable, other nutrients and trans fats may have changed, and the target near 15% of energy from linoleic acid was unusually high. The trial deserves explicit weight, but not category-wide extrapolation.

Table 2 shows the hierarchy rather than forcing a pooled verdict across incompatible trials. The old adverse or null studies widen uncertainty around clinical benefit; they do not align with a broad toxicity signal across modern cohorts and shorter causal trials.

EVIDENCE	DIRECTION	BOUNDARY
PUFA substitution[20]	Events lower	Old trials
Omega-6 trials[21]	Uncertain	Heterogeneous
Recovered trials[25,26]	Null/adverse	Limited generalizability

05 Prospective cohorts and biomarkers weigh against broad cardiovascular toxicity

Prospective dietary studies generally associate higher linoleic-acid intake with neutral or lower coronary and mortality risk. Meta-analyses of cohorts found lower coronary risk when linoleic acid replaced saturated fat or carbohydrate and no excess all-cause mortality across typical intake ranges.[7,27] Reviews of epidemiologic evidence reach the same broad direction, although dietary questionnaires and modeled substitutions remain imperfect.[28] Recent global aggregation of 150 cohorts also weighs against a large adverse association across cardiovascular disease, cancer, and mortality.[29]

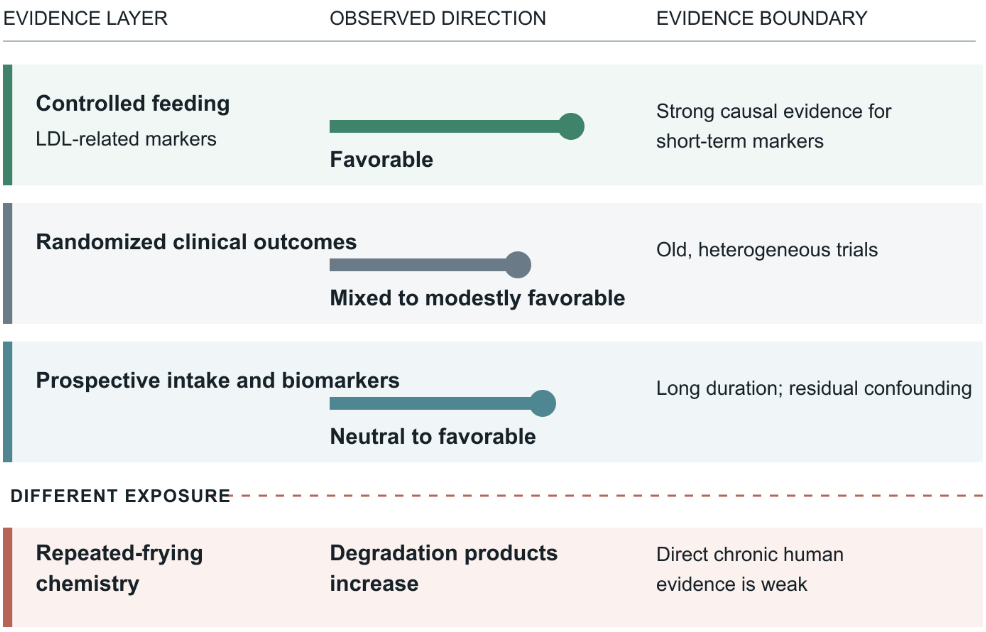
Biomarkers answer a slightly different question. Circulating or tissue fatty acids avoid some recall error but reflect absorption, metabolism, genetics, and background diet. A pooled analysis across 30 prospective studies found higher linoleic-acid biomarkers associated with lower incident cardiovascular risk.[30] The Cardiovascular Health Study and a Swedish population cohort reported neutral-to-favorable associations with cause-specific or total mortality.[31,32] A newer individual-participant analysis across three large cohorts and an updated meta-analysis remained inconsistent with broad harm.[33]

Individual cohorts supply useful boundaries. In 2,837 adults from the Multi-Ethnic Study of Atherosclerosis, neither dietary nor circulating omega-6 PUFA was associated with incident cardiovascular disease over 19,778 person-years, although the study recorded only 189 events.[34] In 2,500 Framingham Offspring participants, erythrocyte omega-6 fatty acids showed little association with cardiovascular outcomes or mortality after adjustment for omega-3 status.[35] A Dutch cohort of about 20,000 adults linked linoleic intake to plasma cholesterol but not a clear ten-year coronary difference.[36] Null results are part of the pattern; they still contradict claims of a large hazard.

Stroke findings are, if anything, more favorable. In a Danish case-cohort study, the highest versus lowest adipose linoleic-acid quartile was associated with lower ischemic stroke risk, hazard ratio 0.78 (95% CI 0.65–0.93), and lower large-artery atherosclerotic stroke risk, 0.61 (0.43–0.88).[37] An individual-participant analysis of UK cohorts found no independent coronary association but lower stroke risk per standard-deviation increase in circulating linoleic acid, odds ratio 0.82 (0.75–0.90).[38] A biomarker meta-analysis and macronutrient-substitution study point in the same direction.[39,8] These associations cannot prove protection, but they are difficult to reconcile with a common, large cardiovascular toxin.

One older Finnish cohort found lower cardiovascular mortality with higher dietary and serum linoleic or total PUFA, while a Southern US cohort reported effect estimates that varied with statin use and other characteristics.[40,41] Such variation is expected when an exposure is embedded in different diets and clinical contexts. The cohort literature therefore supports a boundary: it can rule against large average harm more convincingly than it can quantify an individual benefit.

The convergence across evidence layers is shown in Figure 2. Stronger causal control comes with shorter duration and surrogate outcomes; longer observation comes with more confounding. Their overlap—not any single study—drives the judgment.



LDL = low-density lipoprotein

Figure 2. Human evidence converges against broad toxicity while retaining outcome uncertainty. Controlled feeding provides the clearest causal evidence for LDL improvement; old clinical-event trials are mixed; prospective biomarkers are predominantly neutral or favorable. The repeated-frying chemistry lane is deliberately separated because it describes a different exposure.[21,30,25,42]

06 Human trials do not support a generalized inflammatory effect

The most common mechanistic claim begins with a true biochemical observation: linoleic acid can contribute to membrane pools and downstream lipid mediators. The unsupported leap is that eating more linoleic acid must raise systemic inflammation. A systematic review of randomized trials in healthy people found no evidence that increasing dietary linoleic acid raised standard inflammatory markers.[43] A later meta-analysis likewise found no consistent increase in C-reactive protein, interleukin-6, tumor-necrosis factor, or related blood markers.[44]

Longer-duration randomized evidence remains limited but points in the same direction. A review of omega-3, omega-6, and total PUFA interventions found little evidence that increasing omega-6 worsened inflammatory bowel disease or conventional inflammatory markers, with the caveat that omega-6-specific data were sparse.[45] In adults with abdominal obesity, replacing saturated fat with omega-6 PUFA reduced liver fat and improved lipoproteins without a corresponding inflammatory penalty.[46] Standard biomarkers do not capture every local signaling event, but a clinically meaningful systemic inflammatory effect should eventually leave reproducible human traces.

The metabolism is also more regulated than a simple pathway diagram implies. Dietary linoleic acid is not quantitatively converted into arachidonic acid without constraint, and arachidonic acid can yield both pro-inflammatory and resolving mediators depending on enzymes and context.[47] A systematic review of increasing arachidonic-acid intake in humans did not support a simple generalized inflammatory or health harm.[48] Mechanistic plausibility remains relevant, particularly for genotype or disease subgroups, but it does not overturn the intervention evidence.

07 Glycaemic control and liver fat are neutral to modestly favorable

Randomized evidence does not show that plant-derived PUFA worsens glucose control. A meta-analysis of 13 controlled-feeding studies found essentially no change in fasting glucose, −0.01 mmol/L (95% CI −0.06 to 0.03), while fasting insulin fell 2.6 pmol/L (−4.9 to −0.2) and HOMA-IR fell 0.12 units (−0.23 to −0.01).[49] A larger review of 83 randomized trials found little or no effect of omega-6 or total PUFA on glycaemic outcomes, with limited omega-6-specific evidence.[50] These are small effects, but their direction is incompatible with major glycaemic toxicity.

Liver fat provides a second controlled signal. In overfeeding experiments, excess polyunsaturated and saturated fat produced different patterns of liver and visceral fat accumulation, with saturated fat more strongly promoting ectopic storage.[51] A 12-month trial in 150 adults with prediabetes or type 2 diabetes found that a plant-based high-PUFA diet reduced liver fat by 1.46 percentage points (95% CI −2.42 to −0.51) compared with usual care, although a healthy Nordic diet improved more outcomes and both interventions changed whole dietary patterns.[52] This supports metabolic benefit under specified substitution, not treatment by an isolated oil.

Evidence in established diabetes is heterogeneous. A review of vegetable-oil interventions found six of 12 studies null and six with glycaemic changes, with inconsistent comparators and small samples preventing a firm management claim.[53] In two professional cohorts of people with type 2 diabetes, replacing 2% of energy from saturated fat with linoleic acid was associated with lower cardiovascular mortality, reported hazard ratio 0.85.[54] Prospective meta-analysis also associates dietary or biomarker linoleic acid with neutral or lower diabetes risk, and a Chinese biomarker-derived dietary pattern linked higher unsaturated-fat markers to lower diabetes and coronary disease.[55,56] The trial evidence supports neutrality to small benefit; the cohort evidence cannot establish prevention.

Lipidomics offers a mechanistic bridge. In linked EPIC-Potsdam and DIVAS analyses, 69 plasma lipid features predicted at least one cardiometabolic outcome, and 17 of 19 diet-responsive risk-associated lipids moved favorably when unsaturated fats replaced saturated fats.[57] A secondary analysis likewise found beneficial changes in saturated-fat-containing glycerolipids and sphingolipids, with less clear diabetes associations.[58] These molecular patterns are supportive, but they remain surrogate evidence rather than clinical proof.

08 Cancer evidence is imprecise but does not show a coherent harm signal

Cancer is not one endpoint, and dietary latency makes both trials and cohorts difficult. A meta-analysis of 47 randomized trials lasting at least one year included 108,194 participants, but omega-6-specific cancer evidence came from far fewer people. Its estimate for any cancer was 1.21 (95% CI 0.96–1.53; 4,272 participants), an interval compatible with no effect and with clinically relevant harm.[59] It would be wrong to call that proof of safety.

Prospective evidence is more reassuring but less causal. Across 70 observational reports, high dietary n-6 intake was essentially null for total cancer, risk ratio 1.02 (95% CI 0.99–1.05), while higher blood n-6 was associated with lower risk, 0.92 (0.86–0.98).[60] A broad systematic review found no consistent cancer association across dietary fat categories.[61] Site-specific meta-analyses do not establish higher breast-cancer risk from linoleic acid or a consistent prostate-cancer association from intake or biomarkers.[62,63]

The appropriate conclusion is uncertainty around small site-specific effects, not evidence for a general carcinogen. Cancer trials are underpowered for oil-specific questions, while biomarker cohorts retain confounding and cannot fully represent exposures decades earlier. A long-term claim of zero effect would exceed the evidence, but so does a claim of established cancer harm.

09 Oxidation is real, but repeated high-heat use is the relevant boundary

Polyunsaturated oils are chemically susceptible to lipid peroxidation because double bonds provide reactive sites. Laboratory frying studies detect aldehydes and related products, with greater accumulation in some PUFA-rich oils than in monounsaturated-rich formulations.[64] Temperature and time both increase aldehyde formation in foods during frying.[65] This chemistry establishes a hazard pathway. It does not by itself provide the dose, absorption, detoxification, or disease incidence needed for a human risk estimate.

A systematic review and meta-analysis of 33 heating experiments across 21 oils found little trans-fat formation below 200°C, whereas between 200°C and 240°C total trans fat increased by 0.38 percentage points for each 10°C rise (95% CI 0.20–0.55), with additional effects of prolonged heating.[42] Evidence was sparse above 240°C and heterogeneous across oils and methods. The temperature-response pattern argues against treating fresh salad oil, one controlled sauté, and oil reused through many deep-frying cycles as interchangeable.

Evidence from human exposure is suggestive rather than definitive. Reviews of repeatedly heated oils draw heavily on animal experiments, observational studies, and intermediate cardiovascular markers.[66,67] Cooks exposed to different cooking-oil aerosols showed differences in urinary oxidative-stress biomarkers, but occupational inhalation and uncontrolled workplace conditions do not estimate dietary effects.[68] A small postprandial study in people with diabetes found altered paraoxonase responses after meals containing thermally stressed olive or safflower oil, again an acute surrogate rather than chronic disease.[69]

The usable boundary is shown in Figure 3. Oil should not be heated until it smokes, stored after heavy use, and cycled repeatedly through commercial-style frying. High-oleic formulations, antioxidants, lower temperature, shorter duration, and discarding degraded oil reduce exposure, but no oil becomes immune to thermal damage.

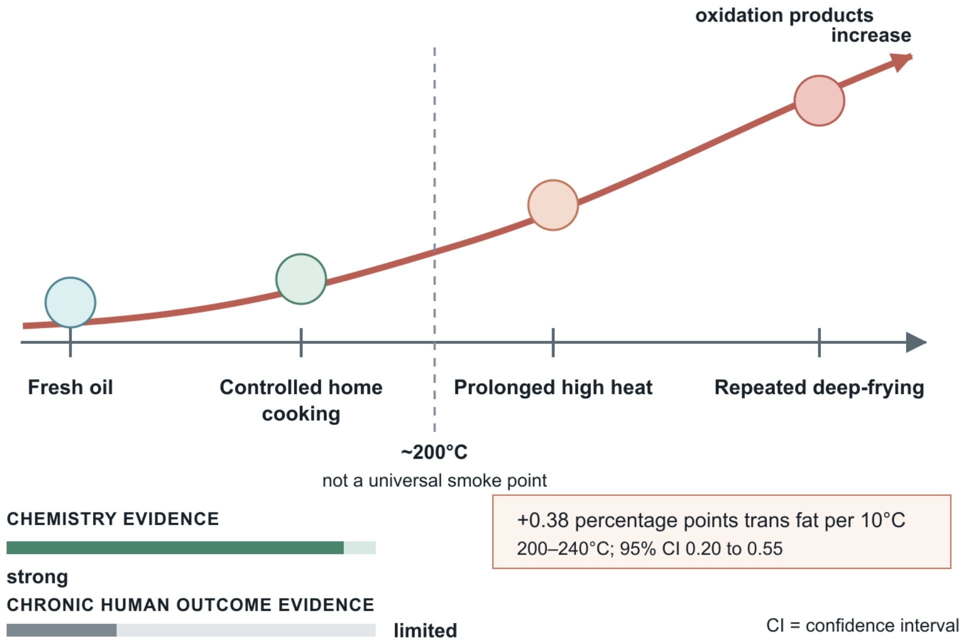


Figure 3. Heating risk follows a gradient rather than the seed-oil label. Chemical degradation rises with temperature, duration, and reuse; the strongest evidence concerns product formation, whereas direct long-term human outcome evidence is weak. The practical boundary is to avoid smoking and repeated reuse, not to equate one controlled cooking episode with industrial frying.[42,64,68,67]

Table 3 separates the questions that are often collapsed into one claim.

EXPOSURE	OBSERVED	GAP
Below 200°C[42]	Little trans fat	Household variation
Repeated frying[64,65]	Aldehydes rise	Human dose–response
Exposed cooks[68]	Biomarkers differ	Mixed exposure routes

10 The omega-6-to-omega-3 ratio can misidentify the harmful component

The omega-6-to-omega-3 ratio compresses two variables into one. A high ratio may mean high omega-6, low omega-3, or both, so its association cannot identify which component matters. In 85,425 UK Biobank participants with 6,461 deaths, the highest versus lowest plasma ratio quintile was associated with higher all-cause mortality, hazard ratio 1.26 (95% CI 1.15–1.38); yet omega-6 and omega-3 concentrations were each independently associated with lower mortality.[70] The ratio looked adverse even though its numerator did not.

This does not mean omega-3 status is irrelevant. Omega-6 and omega-3 fatty acids share parts of their metabolic machinery, and their downstream oxylipins can have different actions.[47] The more informative intervention question is whether omega-3 intake is adequate and what changes when a particular fatty acid is increased—not whether a population meets a universal ratio. A recent randomized trial found that alpha-linolenic and linoleic acids had partly differing effects on lipoprotein traits, again resisting a single ratio score.[71]

11 Food matrix and processing matter beyond the oil molecule

Seed oils are common in ultra-processed foods, but coexistence is not attribution. Fried snacks can combine refined starch, salt, energy density, palatability, repeated heating, and oil. Controlled diets using soybean, canola, corn, or other plant oils do not reproduce that package and often improve lipid markers.[9,10] Therefore, an association between a food pattern and disease cannot be assigned to its oil ingredient without separating these features.

Whole-food comparisons demonstrate the reverse problem. Walnuts and a matched vegetable-oil diet shared an intended unsaturated-fat contrast, yet the walnut diet produced an additional central blood-pressure effect.[17] Prospective substitution of plant for animal foods integrates fiber, micronutrients, processing, and fat quality, not oil alone.[5] A meal is not reducible to one fatty acid.

At the same time, oil-specific trials remain valuable. They isolate a major causal component and show that conventional and high-oleic canola, soybean, corn, sunflower, olive, and camellia oils can differ on selected outcomes.[4,12,10] Recent reviews emphasizing beneficial cardiometabolic effects and reviews framed directly around the seed-oil controversy converge on the need to specify formulation and context.[72,3] The matrix modifies the answer; it does not erase fatty-acid biology.

12 A practical conclusion follows from the evidence hierarchy

For most adults, replacing butter, animal fat, palm-rich fat, or partially hydrogenated fat with an oil richer in unsaturated fatty acids is aligned with the strongest risk-factor evidence and is compatible with modest cardiovascular benefit.[6,20,13] Replacing one unsaturated oil with another is likely to produce a smaller change, and the preferred oil can reasonably depend on flavor, cost, cooking temperature, alpha-linolenic content, and the rest of the diet.[14,1] Adding large quantities without replacing other energy is a different intervention.

Preparation should be treated separately from composition. Fresh oils used in dressings, baking, or controlled stovetop cooking are not equivalent to oil held near or beyond its smoke point and reused repeatedly. The direct human outcome evidence for repeated frying is incomplete, but the chemical gradient justifies avoiding smoking, prolonged holding, and multiple reuse cycles.[42,67] These precautions apply to all cooking fats, with the rate of degradation shaped by unsaturation and antioxidants.

No universal oil mandate follows. People with allergies, specific gastrointestinal disorders, prescribed dietary therapy, or unusual lipid conditions may need individualized advice. At population level, the evidence supports choosing mostly unsaturated fats, maintaining dietary omega-3 sources, and judging packaged foods by the complete product rather than screening for an ingredient name. The claim is modest: ordinary seed-oil use is not a demonstrated broad health hazard.

13 Important uncertainty remains in long-term, oil-specific outcomes

The evidence base has an asymmetry. Short controlled trials answer substitution questions well but rely on surrogate endpoints. Long-term outcome trials are old, heterogeneous, and often poorly documented; contemporary cohorts are large but observational. Even the most comprehensive reviews must combine exposures that differ in oil species, omega-3 content, comparator fat, dose, adherence, and cooking practice.[73,61,1] That heterogeneity limits both accusations of toxicity and declarations of universal benefit.

Biomarker studies improve exposure assessment but introduce metabolic interpretation. Plasma or erythrocyte linoleic acid partly reflects diet, yet it also depends on metabolism, genetics, and other fatty acids; absolute and relative biomarker measures can yield different associations. The repeated neutral-to-inverse cardiovascular pattern is reassuring, but it cannot substitute for randomization.[30,74,75] Recent cohorts extend scale and recency without removing this limitation.[33,29]

The decisive future study would be a modern, adequately powered, long-duration randomized comparison of clearly specified oils in isocaloric, equally processed diets, with verified adherence, cooking exposure, cardiovascular events, diabetes, cancer, and harms. Parallel work should measure aldehyde exposure from household and commercial reuse and test whether FADS genotype or omega-3 status modifies response. Until then, certainty should be highest for LDL lowering, moderate for absence of large average cardiovascular harm, and lower for oil-specific cancer or repeated-frying disease estimates.

14 Conclusion

The cross-cutting pattern is exposure separation. Once the replacement nutrient, oil composition, food matrix, and heating history are distinguished, the apparent conflict becomes narrower: unsaturated-oil substitution improves atherogenic lipoproteins; clinical-event trials allow modest benefit but contain important old outliers; cohorts and biomarkers do not show broad toxicity; and repeated high-heat reuse creates a credible but different chemical hazard. The evidence therefore supports ordinary seed-oil use within a balanced diet more strongly than avoidance of the entire category. What remains unsettled is not whether every seed oil is good or bad, but the size of long-term effects for specific formulations, subgroups, and heating practices. A modern event trial and exposure-calibrated frying studies would resolve those questions more directly than another category-wide argument.

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75 · VERIFIED VIA CROSSREF

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